
Resonant Wire Dipole Design and CST Validation

Analytical Design, Length Tuning, and Far-Field Verification

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Independent Engineering Portfolio Project

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Portfolio Context and Evidence

Project Context. This work originated from a graduate-level antenna engineering simulation project. The present document is an independently prepared public portfolio edition intended to demonstrate analytical antenna design, full-wave electromagnetic simulation, length tuning, radiation-pattern verification, engineering interpretation, and technical documentation.

Public Portfolio Edition. The original academic report has been reorganized and edited for professional and educational use. Course-specific grading material, student-identification information, institutional branding, and administrative content have been removed. This document does not represent an official publication or endorsement by any university, instructor, software vendor, employer, or other institution.

Evidence Classification. The report combines analytical calculations with CST Studio Suite full-wave simulation. Analytical evidence includes wavelength and half-wave estimates, the canonical thin half-wave dipole pattern, and an independent Python calculation of directivity and half-power beamwidth. Simulated evidence includes CST geometry screenshots, reflection-coefficient plots, far-field plots, and the exact marker for the final tuned resonance. No antenna prototype was fabricated, and no calibrated impedance or antenna-range measurements were conducted. Accordingly, the reported results are analytical or simulated evidence, not measured performance.

Simulation Scope. The preserved source archive contains exported CST figures rather than the native CST model or raw numerical curve exports. The final tuned resonance is supported by an exact displayed marker; the preliminary 150 mm and 125 mm resonance values are estimated from plotted curves. Solver version, mesh statistics, stopping criteria, and formal mesh- or boundary-convergence records were not preserved. These limitations are stated explicitly in the numerical-reliability discussion.

Prepared as part of the Hossein Molhem Engineering Portfolio.

Abstract

This report presents the analytical design and CST full-wave simulation of a finite-radius, center-fed wire dipole operating near 1 GHz. A first-order free-space calculation gives a wavelength of 300 mm and an initial half-wave length of 150 mm. The CST model uses a 5 mm-diameter PEC conductor, a 20 mm center gap, a $73\ \Omega$ discrete port, and open-add-space boundaries. Three documented lengths—150 mm, 125 mm, and 135.6 mm—establish the tuning path.

The initial 150 mm model resonated below the 1 GHz target, while shortening the dipole to 125 mm shifted the resonance above the target. The final 135.6 mm model produced a CST marker at 0.99648 GHz with $S_{11,\min} = -47.01186$ dB, corresponding to a positive return loss of 47.01186 dB relative to the $73\ \Omega$ port reference. The frequency deviation from 1 GHz is 3.52 MHz, or 0.352%.

The final CST far-field result displays the expected broadside toroidal pattern with axial nulls and a peak directivity of 2.149 dBi. Independent evaluation of the canonical thin half-wave dipole pattern gives 2.15 dBi directivity and a 78.08° half-power beamwidth, in close agreement with the displayed CST pattern. The study therefore connects closed-form antenna theory, parameterized full-wave modeling, resonance control through length tuning, and analytical-to-simulation verification.

All performance results are analytical or simulated. No fabricated prototype or measured antenna data are claimed. The absence of the native CST model, raw exports, and formal mesh- and boundary-convergence records limits the conclusions to the preserved evidence package.

Project Objective

The objective of this project is to establish a traceable antenna-engineering workflow for a center-fed resonant wire dipole operating near 1 GHz. The work connects a closed-form half-wave baseline to a parameterized CST model, demonstrates resonance control through physical-length tuning, and verifies the final radiation pattern against canonical dipole theory.

The project has four principal technical objectives:

1. Calculate the free-space wavelength and first-order half-wave dipole length at the 1 GHz design frequency.
2. Construct the finite-radius, center-fed dipole model in CST Studio Suite and document the conductor geometry, center gap, discrete-port reference, boundary condition, frequency sweep, and far-field monitors.
3. Tune the total dipole length through documented 150 mm, 125 mm, and 135.6 mm cases, and quantify the final resonant frequency and reflection coefficient.
4. Compare the final CST far-field behavior with an independent analytical calculation of the thin half-wave dipole directivity and half-power beamwidth.

A secondary objective is to distinguish exact marker values, displayed figure values, and plot-derived estimates, while documenting the numerical and experimental evidence still required before a hardware-performance claim would be justified.

The final outcome is intended to provide a concise, university-style engineering report that remains suitable for a public technical portfolio. All reported performance results are analytical or simulated; no fabricated prototype or measured data are claimed.

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Chapter 1

Resonant Wire Dipole Design and CST Validation

1.1 Project Overview

1.1.1 Engineering objective

The engineering problem is to tune a practical finite-radius wire dipole so that its input match occurs near 1 GHz, while retaining the expected broadside radiation pattern of a resonant half-wave dipole. The project is intentionally framed as a compact verification case: analytical theory supplies a canonical baseline, CST provides the full-wave response, and the comparison identifies both agreement and missing evidence.

1.1.2 Design requirements and configuration

Table 1.1: Design requirements and documented model parameters.

Item	Documented value or requirement
Target frequency	1.000 GHz
Free-space wavelength used for design	300 mm
Initial theoretical length	150 mm
Final tuned total length	135.6 mm
Conductor model	PEC cylinder, 5 mm diameter
Center feed gap	20 mm
Port reference impedance	$73\ \Omega$
Frequency sweep	0.5 GHz to 2.0 GHz
Far-field monitors	0.9 GHz and 1.0 GHz
Primary outputs	S_{11} , 2D pattern, 3D directivity pattern

1.1.3 Key engineering outcome

The final 135.6 mm model produced the exact CST marker result $f_{\text{res}} = 0.996\ 48\ \text{GHz}$ and $S_{11,\text{min}} = -47.011\ 86\ \text{dB}$. Its displayed directivity at 1 GHz was 2.149 dBi, essentially equal to the analytical half-wave value. These results support a tuned resonant dipole interpretation under the documented $73\ \Omega$ reference condition.

1.1.4 Reporting decision

The marker-based final run is used as the quantitative design result. The earlier 150 mm and 125 mm cases are treated as directional tuning evidence because their numerical minima are available only from plotted curves, not raw exported data.

1.2 Scope and Evidence

1.2.1 Evidence classes

Analytical evidence. The analytical record includes wavelength, ideal half-wave length, the thin resonant half-wave dipole pattern, numerical directivity integration, and analytical HPBW.

Simulated evidence. The simulation record includes CST geometry screenshots, three length-tuning S_{11} plots, and two-dimensional and three-dimensional far-field figures. The final S_{11} figure contains an exact marker at 0.996 48 GHz.

Measured evidence. No prototype, VNA calibration, balun, feed cable, chamber range, gain-transfer test, or uncertainty budget is included. No measured claim is made.

1.2.2 Public-edition boundaries

Administrative course information and student identifiers were removed. Native CST binary models were not present in the supplied archive and are therefore not included. The public package preserves the supplied figures, an independent calculation script, a run manifest, and explicit numerical limitations.

1.2.3 Claim discipline

The report distinguishes among three levels of numerical confidence:

1. **Exact exported marker:** final resonance and final minimum S_{11} .
2. **Displayed figure value:** far-field directivity and angular-width annotations.
3. **Plot-derived estimate:** resonance and minimum depth for the two preliminary lengths.

This distinction prevents screenshot estimates from being presented with unsupported precision.

1.3 Theory and Analytical Design

1.3.1 Wavelength and first-order length

Using the design value $c \approx 3 \times 10^8$ m/s,

$$\lambda_0 = \frac{c}{f_0} = \frac{3 \times 10^8 \text{ m/s}}{1 \times 10^9 \text{ Hz}} = 0.300 \text{ m} = 300 \text{ mm.} \quad (1.1)$$

The first-order half-wave length is therefore

$$L_0 = \frac{\lambda_0}{2} = 150 \text{ mm.} \quad (1.2)$$

This is an analytical starting point rather than a guarantee of resonance for a finite-diameter, gapped, port-excited model.

1.3.2 Canonical resonant-dipole pattern

For a thin, center-fed resonant half-wave dipole aligned with the z -axis, the normalized far-field magnitude pattern is approximated by [1]

$$F(\theta) = \left| \frac{\cos\left(\frac{\pi}{2} \cos \theta\right)}{\sin \theta} \right|. \quad (1.3)$$

The maximum occurs at $\theta = 90^\circ$ and the axial nulls occur at $\theta = 0^\circ$ and 180° . Numerical integration of the power pattern gives

$$D = 1.6409 = 2.15 \text{ dBi}, \quad (1.4)$$

and the -3 dB points give an HPBW of 78.08° .

1.3.3 Reflection coefficient and return loss

For an antenna impedance Z_A referenced to the CST port impedance Z_0 ,

$$\Gamma = \frac{Z_A - Z_0}{Z_A + Z_0}, \quad S_{11,\text{dB}} = 20 \log_{10} |\Gamma|. \quad (1.5)$$

Return loss is conventionally positive:

$$\text{RL} = -20 \log_{10} |\Gamma| = -S_{11,\text{dB}}. \quad (1.6)$$

Thus the final plotted value $S_{11} = -47.01186 \text{ dB}$ corresponds to a return loss of 47.01186 dB , specifically under the 73Ω reference used in the CST port.

1.3.4 Independent analytical calculation

The script in Appendix A evaluates Eq. (1.3), finds the half-power crossings, integrates beam solid angle, and records the comparison values independently of CST.

1.4 Model and Simulation Method

1.4.1 CST geometry

The model is a center-fed PEC cylindrical dipole on the z -axis. A centered vacuum cylinder creates a 20 mm gap. A discrete face port spans the inner circular faces and uses $Z_0 = 73 \Omega$. Open-add-space boundaries are documented in the supplied report

source. The construction follows the wire-dipole tutorial workflow identified in the original project record [2].

For the final total length $L = 135.6$ mm, each metal arm is

$$L_{\text{arm}} = \frac{L - \text{gap}}{2} = \frac{135.6 - 20}{2} = 57.8 \text{ mm.} \quad (1.7)$$

The documented arm coordinates are $z = -67.8$ mm to -10 mm and $z = +10$ mm to $+67.8$ mm.

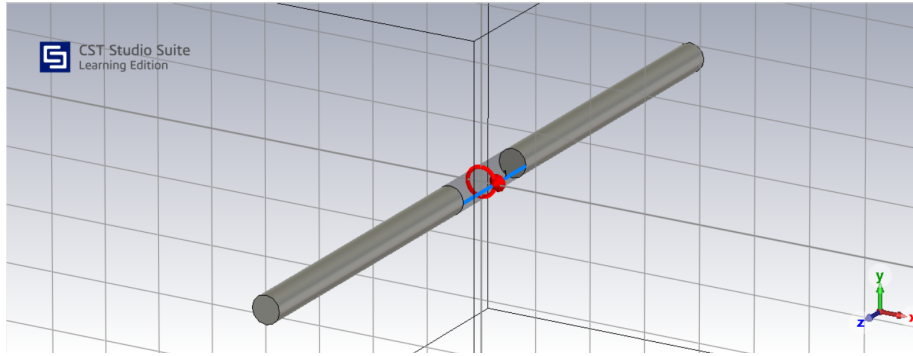


Figure 1.1: Final CST geometry for the 135.6 mm tuned model (simulated evidence).

1.4.2 Length-tuning sequence

Table 1.2: Documented length-tuning sequence.

Run ID	Total length	Purpose	Evidence status
SIM-150	150.0 mm	Analytical half-wave starting point	Plot-derived resonance estimate; 2D and 3D patterns at 0.9 GHz and 1.0 GHz
SIM-125	125.0 mm	Confirm upward resonance shift after shortening	Plot-derived resonance estimate
SIM-136	135.6 mm	Final tuned model near 1 GHz	Exact S_{11} marker; 2D and 3D directivity plots

1.4.3 Available and unavailable solver records

The supplied archive preserves the sweep band, geometry, materials, port reference, boundary description, monitor frequencies, and result figures. It does not preserve the

native CST model, software build number, solver type, mesh statistics, convergence history, or raw exported curves.

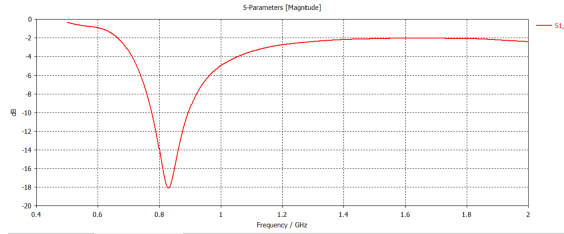
1.4.4 Numerical qualification

The model can be interpreted and partially reconstructed from the documented geometry, but the exact numerical run cannot be reproduced bit-for-bit without the native model and solver records. Consequently, this report verifies the preserved evidence package; it does not claim a completed mesh- or boundary-convergence campaign.

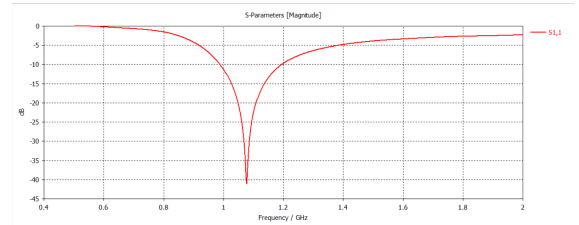
1.5 Results and Verification

1.5.1 Length tuning and resonance control

The 150 mm analytical starting length resonated below 1 GHz; the plotted minimum is approximately 0.83 GHz with $S_{11} \approx -18$ dB. Shortening the model to 125 mm moved the plotted minimum above the target to approximately 1.08 GHz with $S_{11} \approx -40$ dB. These two results establish the expected inverse relationship between physical length and resonant frequency.



(a) SIM-150, plot-derived minimum near 0.83 GHz.



(b) SIM-125, plot-derived minimum near 1.08 GHz.

Figure 1.2: Preliminary length-tuning evidence from CST. Numerical values are approximate because raw curve exports were not supplied.

1.5.2 Final tuned match

For SIM-136, the CST marker gives

$$f_{\text{res}} = 0.99648 \text{ GHz}, \quad S_{11,\text{min}} = -47.01186 \text{ dB}. \quad (1.8)$$

The absolute frequency error is

$$\Delta f = |1.00000 - 0.99648| = 0.00352 \text{ GHz} = 3.52 \text{ MHz}, \quad (1.9)$$

and the relative error is

$$\frac{\Delta f}{f_0} \times 100 = 0.352\%. \quad (1.10)$$

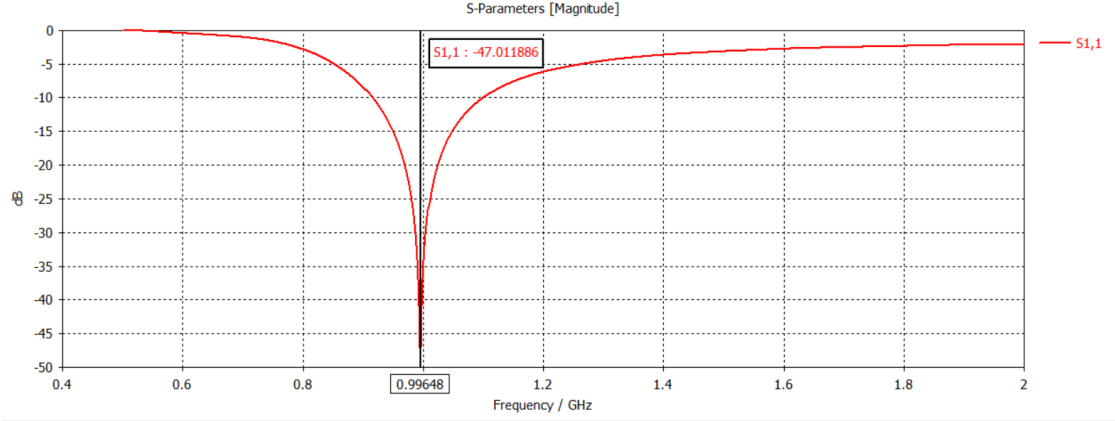
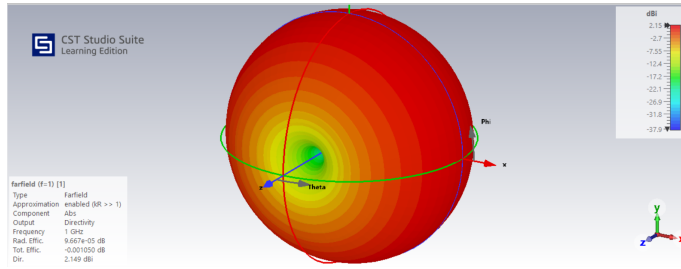


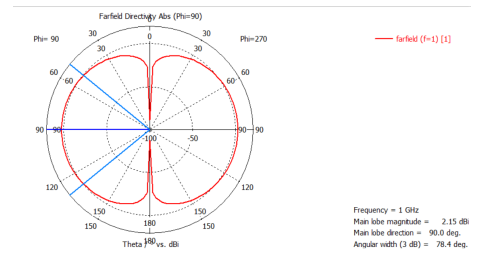
Figure 1.3: Exact CST marker for SIM-136: $S_{11} = -47.01186$ dB at 0.99648 GHz. The equivalent positive return loss is 47.01186 dB relative to $73\ \Omega$.

1.5.3 Far-field behavior

At 1 GHz, the final 3D plot displays a broadside toroidal pattern with axial nulls and a peak directivity of 2.149 dBi. The associated $\phi = 90^\circ$ cut displays a main-lobe magnitude of approximately 2.15 dBi and a -3 dB angular width of approximately 78.4° .



(a) 3D directivity at 1 GHz.



(b) $\phi = 90^\circ$ principal-plane cut.

Figure 1.4: Final CST far-field evidence for SIM-136.

1.5.4 Analytical verification

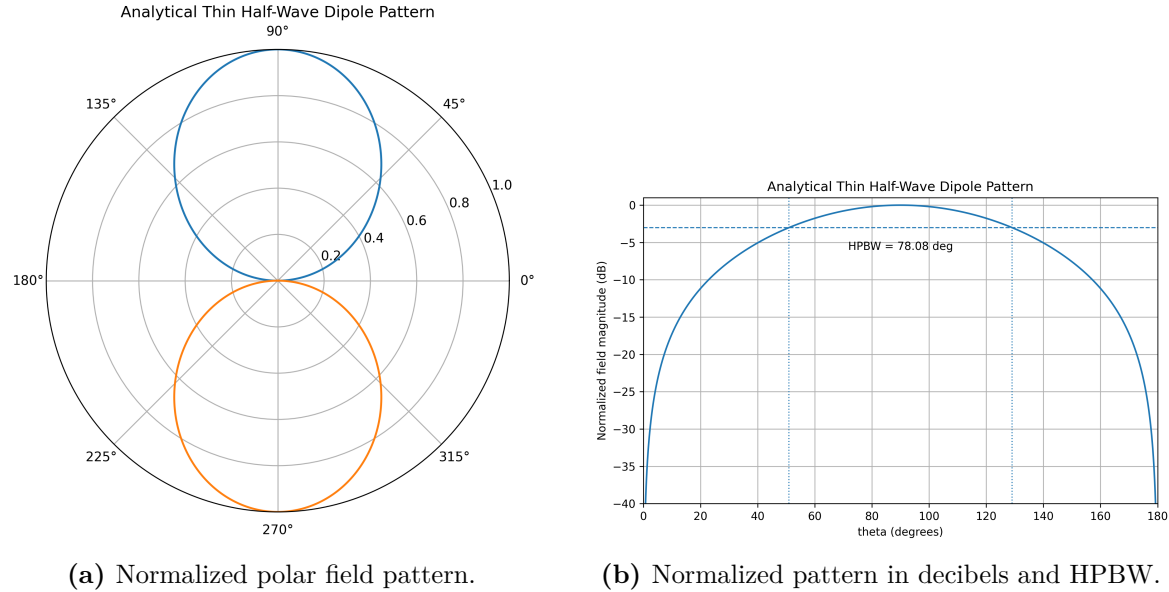


Figure 1.5: Independent analytical Python evaluation of the canonical thin half-wave dipole pattern.

Table 1.3: Comparison of analytical and final simulated radiation evidence.

Metric	Analytical	Final CST display	Interpretation
Reference frequency	1.000 GHz	Pattern at 1.000 GHz	Same comparison frequency
Peak directivity	2.15 dBi	2.149 dBi	Agreement within displayed precision
HPBW	78.08°	approximately 78.4°	Difference about 0.32°
Pattern topology	Broadside, axial nulls	Broadside, axial nulls	Canonical dipole behavior

1.5.5 Verification and validation status

The analytical script independently verifies the canonical pattern metrics. The CST figures verify the qualitative tuning trend and final simulated response. Experimental validation is absent; therefore, the package demonstrates analytical-to-simulation agreement rather than hardware performance.

1.6 Engineering Discussion

1.6.1 Tuned Length Versus the Half-Wave Estimate

The first-order value 150 mm assumes an idealized thin resonant dipole. The CST model includes a finite 5 mm conductor diameter, a large 20 mm center gap, and a discrete-port field region. These features change the current distribution and terminal reactance, so the resonant physical length need not equal exactly $\lambda_0/2$. The documented tuning sequence found 135.6 mm, or approximately $0.452\lambda_0$, for the selected model and port definition.

1.6.2 Meaning of the Deep Reflection-Coefficient Minimum

The -47.01 dB marker is evidence of a very close match to the chosen 73Ω reference at one frequency. It is not evidence of broadband behavior, and it does not establish the match that would be obtained in a 50Ω system. No impedance curve or raw data were provided to quantify resistance and reactance separately.

1.6.3 Interpretation of the matching result

The final result is reported as a narrowband tuned match under a 73Ω reference. It should not be generalized to a 50Ω feed system without rerunning or renormalizing the model and documenting the resulting impedance match.

1.6.4 Numerical reliability

The tuning trend is physically plausible and monotonic across the three documented lengths. The final marker and canonical pattern agreement increase confidence in the model interpretation. However, the following checks are not available:

- mesh refinement or cell-count convergence;
- open-boundary spacing sensitivity;
- port-gap and port-definition sensitivity;
- raw-data re-extraction of resonance, bandwidth, and HPBW;
- comparison with a finite-conductivity conductor;
- independent solver or measurement validation.

1.6.5 Qualification of the directivity comparison

The difference between analytical and CST directivity is smaller than the displayed precision and should not be interpreted as a metrologically significant error measure. The comparison supports the expected pattern class, not sub-hundredth-decibel accuracy.

1.6.6 Industrial relevance and next step

This compact workflow mirrors a common antenna-development sequence: establish a closed-form starting geometry, parameterize the electromagnetic model, bracket the resonance through a coarse sweep, select a tuned configuration, and verify pattern behavior. For a production-oriented continuation, the next defensible steps are a $50\ \Omega$ feed design or balun, mesh and boundary convergence, raw-data export, finite-conductivity analysis, fabrication, calibrated VNA measurement, and controlled far-field testing.

1.7 Engineering Verification Matrix

The matrix links each public claim to its evidence class, source, and qualification.

Table 1.4: Engineering verification matrix.

ID	Public claim	Evidence class	Evidence location	Limitation or qualification
V1	The first-order half-wave length at 1 GHz is 150 mm.	Analytical	Eqs. (1.1) and (1.2); Appendix A	Uses $c \approx 3 \times 10^8$ m/s and an idealized first-order model.
V2	Shortening the dipole shifts the resonance upward.	Simulated	Fig. 1.2; run manifest SIM-150 and SIM-125	Preliminary values are plot-derived estimates, not raw-data extractions.
V3	The final resonance is 0.996 48 GHz.	Simulated	Fig. 1.3; run SIM-136	Exact value comes from the displayed CST marker.

Continued on next page

ID	Public claim	Evidence class	Evidence location	Limitation or qualification
V4	The final minimum is $S_{11} = -47.01186$ dB relative to 73Ω .	Simulated	Fig. 1.3; run SIM-136	A narrowband match to 73Ω ; not a 50Ω qualification.
V5	The final frequency error is 0.352 %.	Analytical + Simulated	Results chapter; verification record	Calculated from the exact displayed marker and the 1 GHz target.
V6	The final pattern is broadside with axial nulls.	Simulated	Fig. 1.4	Based on exported far-field figures; no raw angular export.
V7	The final displayed directivity is 2.149 dBi.	Simulated	3D figure in Fig. 1.4	Displayed precision exceeds what can be independently validated without raw data.
V8	The analytical directivity and HPBW are 2.15 dBi and 78.08° .	Analytical	Fig. 1.5; Appendix A	Thin resonant half-wave dipole approximation.
V9	Analytical and simulated pattern metrics agree closely.	Analytical + Simulated	Table 1.3	Supports canonical pattern behavior, not experimental validation.
V10	No measured hardware performance is claimed.	Measured status	Portfolio Context and Evidence; validation-status section	Fabrication and calibrated testing are future work.
V11	Formal numerical convergence was not demonstrated.	Qualified status	Numerical-reliability discussion; model record	Mesh, boundary, solver, and port sensitivity records were absent from the supplied archive.

1.8 Conclusion

A finite-radius PEC wire dipole was analytically initialized and numerically tuned for operation near 1 GHz. The first-order half-wave value of 150 mm resonated below the target in the documented CST model, while 125 mm resonated above it. The final 135.6 mm configuration produced an exact CST marker at 0.996 48 GHz, only 0.352 % below the target, with $S_{11} = -47.011\ 86$ dB relative to the selected $73\ \Omega$ reference.

The final far-field evidence shows the expected broadside toroidal pattern and axial nulls. The displayed directivity of 2.149 dBi and angular width of approximately 78.4° agree closely with the analytical thin half-wave values of 2.15 dBi and 78.08° . The verified conclusion is therefore that the preserved CST model behaves as a tuned resonant wire dipole and reproduces canonical half-wave radiation behavior.

The evidence remains analytical and simulated. A complete engineering qualification would require the native model, raw exports, mesh and boundary convergence, a practical feed or balun, fabrication, calibrated impedance measurement, and controlled far-field testing.

Appendix A

Analytical Verification Script

The following script independently computes the thin half-wave dipole pattern, directivity, HPBW, frequency error, and analytical figures used in this report.

```

1  #!/usr/bin/env python3
2  """Independent analytical verification for the resonant wire dipole portfolio."""
3  from __future__ import annotations
4
5  from pathlib import Path
6  import math
7  import numpy as np
8  import matplotlib.pyplot as plt
9
10 C0 = 3.0e8
11 DESIGN_FREQUENCY_HZ = 1.0e9
12 FINAL_LENGTH_MM = 135.6
13 CST_RESONANCE_GHZ = 0.99648
14 CST_S11_DB = -47.01186
15 CST_DIRECTIVITY_DBI = 2.149
16
17 ROOT = Path(__file__).resolve().parents[1]
18 FIG_DIR = ROOT / "figures" / "analytical"
19 FIG_DIR.mkdir(parents=True, exist_ok=True)
20
21
22 def trapz(y: np.ndarray, x: np.ndarray) -> float:
23     if hasattr(np, "trapezoid"):
24         return float(np.trapezoid(y, x))
25     return float(np.trapz(y, x))
26
27
28 def normalized_pattern(theta: np.ndarray) -> np.ndarray:
29     sin_theta = np.sin(theta)
30     numerator = np.cos(0.5 * np.pi * np.cos(theta))
31     field = np.zeros_like(theta)
32     mask = np.abs(sin_theta) > 1.0e-12

```

```

33     field[mask] = np.abs(numerator[mask] / sin_theta[mask])
34     return field / np.max(field)
35
36
37 def half_power_angles(theta: np.ndarray, field: np.ndarray) -> tuple[float, float,
float]:
38     power = field**2
39     peak = int(np.argmax(power))
40     left_i = np.where(power[:peak] <= 0.5)[0][-1]
41     right_i = np.where(power[peak:] <= 0.5)[0][0] + peak
42     left = np.interp(0.5, [power[left_i], power[left_i + 1]], [theta[left_i],
theta[left_i + 1]])
43     right = np.interp(0.5, [power[right_i], power[right_i - 1]], [theta[right_i],
theta[right_i - 1]])
44     return math.degrees(left), math.degrees(right), math.degrees(right - left)
45
46
47 def directivity(theta: np.ndarray, field: np.ndarray) -> tuple[float, float]:
48     omega_a = 2.0 * np.pi * trapz(field**2 * np.sin(theta), theta)
49     d_linear = 4.0 * np.pi / omega_a
50     return d_linear, 10.0 * math.log10(d_linear)
51
52
53 def save_plots(theta: np.ndarray, field: np.ndarray, left: float, right: float,
hpbw: float) -> None:
54     fig = plt.figure(figsize=(6, 6))
55     ax = fig.add_subplot(111, projection="polar")
56     ax.plot(theta, field)
57     ax.plot(2.0 * np.pi - theta, field)
58     ax.set_title("Analytical Thin Half-Wave Dipole Pattern")
59     ax.set_rmax(1.0)
60     ax.grid(True)
61     fig.tight_layout()
62     fig.savefig(FIG_DIR / "analytical_half_wave_dipole_polar.png", dpi=300,
bbox_inches="tight")
63     plt.close(fig)
64
65     field_db = 20.0 * np.log10(np.maximum(field, 1.0e-5))
66     fig = plt.figure(figsize=(7.5, 4.8))
67     plt.plot(np.degrees(theta), field_db)
68     plt.axhline(-3.0, linestyle="--", linewidth=1.0)
69     plt.axvline(left, linestyle=":", linewidth=1.0)
70     plt.axvline(right, linestyle=":", linewidth=1.0)
71     plt.text(90.0, -6.0, f"HPBW = {hpbw:.2f} deg", ha="center")
72     plt.xlim(0, 180)
73     plt.ylim(-40, 1)
74     plt.xlabel("theta (degrees)")
75     plt.ylabel("Normalized field magnitude (dB)")
76     plt.title("Analytical Thin Half-Wave Dipole Pattern")
77     plt.grid(True)
78     fig.tight_layout()
79     fig.savefig(FIG_DIR / "analytical_half_wave_dipole_pattern_db.png", dpi=300,

```

```

80     bbox_inches="tight")
81     plt.close(fig)
82
83 def main() -> None:
84     wavelength_m = C0 / DESIGN_FREQUENCY_HZ
85     theta = np.linspace(0.0, np.pi, 20001)
86     field = normalized_pattern(theta)
87     left, right, hpbw = half_power_angles(theta, field)
88     d_linear, d_dbi = directivity(theta, field)
89     error_mhz = abs(1.0 - CST_RESONANCE_GHZ) * 1000.0
90     error_pct = abs(1.0 - CST_RESONANCE_GHZ) * 100.0
91     save_plots(theta, field, left, right, hpbw)
92
93     print("Resonant Wire Dipole Analytical Verification")
94     print(f"Design frequency: {DESIGN_FREQUENCY_HZ/1e9:.3f} GHz")
95     print(f"Wavelength: {wavelength_m*1000:.3f} mm")
96     print(f"First-order half-wave length: {wavelength_m*500:.3f} mm")
97     print(f"Final CST length: {FINAL_LENGTH_MM:.1f} mm")
98     print(f"Analytical half-power angles: {left:.2f} deg, {right:.2f} deg")
99     print(f"Analytical HPBW: {hpbw:.2f} deg")
100    print(f"Analytical directivity: {d_linear:.4f} = {d_dbi:.2f} dBi")
101    print(f"Final CST resonance: {CST_RESONANCE_GHZ:.5f} GHz")
102    print(f"Final CST S11: {CST_S11_DB:.5f} dB")
103    print(f"Equivalent return loss: {-CST_S11_DB:.5f} dB")
104    print(f"Final CST directivity: {CST_DIRECTIVITY_DBI:.3f} dBi")
105    print(f"Frequency error: {error_mhz:.2f} MHz = {error_pct:.3f} percent")
106
107
108 if __name__ == "__main__":
109     main()

```

Listing A.1: Analytical verification script for the resonant wire dipole.

```

Resonant Wire Dipole Analytical Verification
Design frequency: 1.000 GHz
Wavelength: 300.000 mm
First-order half-wave length: 150.000 mm
Final CST length: 135.6 mm
Analytical half-power angles: 50.96 deg, 129.04 deg
Analytical HPBW: 78.08 deg
Analytical directivity: 1.6409 = 2.15 dBi
Final CST resonance: 0.99648 GHz
Final CST S11: -47.01186 dB
Equivalent return loss: 47.01186 dB
Final CST directivity: 2.149 dBi
Frequency error: 3.52 MHz = 0.352 percent

```

Listing A.2: Recorded output from the analytical verification script.

Appendix B

Simulation and Model Record

Table B.1: Public simulation-model record.

Field	Preserved record
Software	CST Studio Suite Learning Edition
Geometry	PEC cylindrical center-fed dipole, 5 mm diameter, 20 mm gap
Port	Discrete face port, $73\ \Omega$ reference
Boundary	Open-add-space in all directions, as documented in supplied source
Sweep	0.5 GHz to 2.0 GHz
Far-field monitors	0.9 GHz and 1.0 GHz
Length cases	150.0 mm, 125.0 mm, 135.6 mm
Exact marker	0.996 48 GHz, $S_{11} = -47.011\ 86\ \text{dB}$
Unavailable	Native model, solver build/type, mesh statistics, convergence history, raw CSV exports

The machine-readable run manifest is stored at `simulation_records/run_manifest.csv`. The screenshot file names in that manifest are retained unchanged from the supplied project archive to preserve traceability.

References

- [1] W. L. Stutzman and G. A. Thiele, *Antenna Theory and Design*, 3rd ed. John Wiley & Sons, 2012.
- [2] *Wire dipole antenna tutorial*, CST Studio Suite learning resource used in the original project workflow, Dassault Systèmes SIMULIA.